

Online Resource 1

Supplementary methods and results

Evaluating Jev for biomedical evidence screening: discrimination, calibration and high-recall workload

Health Information Science and Systems

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Contents: Supplementary methods; Tables S1-S10; interpretation and data dictionary. These materials report frozen analyses, not additional model fitting. Online Resource 2 provides complete-precision CSV values and the integrity audit. Displayed decimal results are rounded to four places. NA denotes unavailable or inapplicable values, not zero.

Supplementary methods

The analysis unit is a review-record pair. Labels are eventual full-text inclusion, not independently adjudicated title/abstract screening decisions. The benchmark contains 17,191 usable records, 392 final inclusions and ten selected reviews. The same publication can occur in multiple reviews.

Primary pooled scores are native probabilities for Jev and feature-based leave-one-review-out logistic regression (LR), and within-review rank percentiles for retrieval baselines. All-method percentile and review-macro analyses address this scale difference. LR is fitted on ten engineered predictors, not high-dimensional text vectors. Removing cross-review overlaps in a sensitivity subset does not retrain LR or establish publication-disjoint validation.

Paired within-review row bootstrap: 5,000 replicates, seed 20260921, fixed predictions. Primary intervals are percentile 95% intervals. The exploratory Bonferroni intervals have 99.375% coverage for eight contrasts. Review resampling uses 10,000 replicates across ten reviews. These procedures do not include model-training variability.

The high-recall target is $\text{ceil}(0.95 \times \text{final inclusions})$ per review. Primary workload includes the full boundary tie. WSS@95 equals 0.95 minus screening fraction; achieved recall is reported separately. Under random tie ordering, expected screened count equals $B + r(M + 1)/(Q + 1)$, with B records above the boundary, M tied records, Q tied inclusions and r additional inclusions required. This label-informed expectation is not a prospective stopping rule.

The unchanged inference prompt and hash are reproduced in manuscript Appendix A. Frozen analysis environment: Python 3.12.14, NumPy 2.3.5 and pandas 3.0.1. These version values describe the archived analysis, not the software used to typeset this supplement. Full inference and comparator dependency lockfiles are not supplied in these Online Resources.

Supplementary discrimination results

Table S1 Review-specific discrimination

Source: per_review_metrics.csv. AP is non-interpolated average precision. Values are review-specific point estimates; no per-review confidence intervals are asserted.

Review	Method	Records	Inclusions	AUROC	AP
Bos_2018	Jev	4878	10	0.9461	0.4376
Chou_2003	Jev	1907	15	0.9944	0.6859
Donners_2021	Jev	258	15	0.9422	0.6440
Meijboom_2021	Jev	882	37	0.8652	0.3731
Menon_2022	Jev	975	74	0.9365	0.7345
Nelson_2002	Jev	366	80	0.8615	0.6715
Oud_2018	Jev	952	20	0.9449	0.3221
Wolters_2018	Jev	4278	19	0.9412	0.4383
van_der_Valk_2021	Jev	725	89	0.7821	0.5114
van_der_Waal_2022	Jev	1970	33	0.9663	0.3858
Bos_2018	BM25	4878	10	0.8749	0.0174
Chou_2003	BM25	1907	15	0.8751	0.0523
Donners_2021	BM25	258	15	0.6263	0.1479
Meijboom_2021	BM25	882	37	0.8135	0.1466
Menon_2022	BM25	975	74	0.6026	0.1011
Nelson_2002	BM25	366	80	0.5772	0.2526
Oud_2018	BM25	952	20	0.8531	0.0788
Wolters_2018	BM25	4278	19	0.7469	0.0103
van_der_Valk_2021	BM25	725	89	0.6500	0.1870
van_der_Waal_2022	BM25	1970	33	0.6867	0.0428
Bos_2018	TF-IDF	4878	10	0.9649	0.0412
Chou_2003	TF-IDF	1907	15	0.8220	0.0468
Donners_2021	TF-IDF	258	15	0.7021	0.2333
Meijboom_2021	TF-IDF	882	37	0.7004	0.0720

Review	Method	Records	Inclusions	AUROC	AP
Menon_2022	TF-IDF	975	74	0.6830	0.1256
Nelson_2002	TF-IDF	366	80	0.5273	0.2502
Oud_2018	TF-IDF	952	20	0.7917	0.0675
Wolters_2018	TF-IDF	4278	19	0.9305	0.0831
van_der_Valk_2021	TF-IDF	725	89	0.6805	0.2268
van_der_Waal_2022	TF-IDF	1970	33	0.8232	0.0787
Bos_2018	MiniLM	4878	10	0.9010	0.0131
Chou_2003	MiniLM	1907	15	0.8783	0.0535
Donners_2021	MiniLM	258	15	0.8560	0.3242
Meijboom_2021	MiniLM	882	37	0.8231	0.1385
Menon_2022	MiniLM	975	74	0.8461	0.2858
Nelson_2002	MiniLM	366	80	0.7119	0.3600
Oud_2018	MiniLM	952	20	0.7617	0.0494
Wolters_2018	MiniLM	4278	19	0.9733	0.1638
van_der_Valk_2021	MiniLM	725	89	0.8223	0.4118
van_der_Waal_2022	MiniLM	1970	33	0.8896	0.1078
Bos_2018	LR	4878	10	0.8039	0.0082
Chou_2003	LR	1907	15	0.8364	0.0543
Donners_2021	LR	258	15	0.7394	0.1730
Meijboom_2021	LR	882	37	0.8189	0.1288
Menon_2022	LR	975	74	0.6851	0.1211
Nelson_2002	LR	366	80	0.5664	0.2673
Oud_2018	LR	952	20	0.7495	0.0461
Wolters_2018	LR	4278	19	0.7858	0.0134
van_der_Valk_2021	LR	725	89	0.7345	0.2609
van_der_Waal_2022	LR	1970	33	0.7724	0.0783

Paired discrimination comparisons

Table S2 Paired differences and uncertainty

Source: paired_differences.csv. Differences are Jev minus comparator. AUPRC_AP means average precision. Intervals condition on the selected reviews and fixed predictions; multiplicity-adjusted intervals are exploratory.

Score scale	Comparison	Metric	Difference	95% lower	95% upper	99.375% lower	99.375% upper
primary	Jev - BM25	AUROC	0.2564	0.2266	0.2881	0.2140	0.3004
primary	Jev - BM25	AUPRC_AP	0.4607	0.4082	0.5143	0.3869	0.5342
primary	Jev - TF-IDF	AUROC	0.2338	0.2048	0.2641	0.1917	0.2776
primary	Jev - TF-IDF	AUPRC_AP	0.4504	0.3985	0.5034	0.3755	0.5237
primary	Jev - MiniLM	AUROC	0.1218	0.0979	0.1456	0.0884	0.1555
primary	Jev - MiniLM	AUPRC_AP	0.4183	0.3659	0.4707	0.3463	0.4927
primary	Jev - LR	AUROC	0.1520	0.1222	0.1815	0.1114	0.1936
primary	Jev - LR	AUPRC_AP	0.4164	0.3635	0.4696	0.3413	0.4864
all_models_rank_pct	Jev - BM25	AUROC	0.1905	0.1537	0.2295	0.1365	0.2435
all_models_rank_pct	Jev - BM25	AUPRC_AP	0.2537	0.2123	0.3025	0.1957	0.3171
all_models_rank_pct	Jev - TF-IDF	AUROC	0.1679	0.1320	0.2049	0.1166	0.2180
all_models_rank_pct	Jev - TF-IDF	AUPRC_AP	0.2434	0.2023	0.2886	0.1879	0.3028
all_models_rank_pct	Jev - MiniLM	AUROC	0.0559	0.0231	0.0875	0.0062	0.0996
all_models_rank_pct	Jev - MiniLM	AUPRC_AP	0.2113	0.1707	0.2561	0.1567	0.2709
all_models_rank_pct	Jev - LR	AUROC	0.1551	0.1184	0.1926	0.0993	0.2035
all_models_rank_pct	Jev - LR	AUPRC_AP	0.2476	0.2061	0.2955	0.1891	0.3100

Supplementary high-recall workload

Table S3a Workload with complete boundary ties

Source: screening_per_review.csv. Fractions are on the 0-1 scale. WSS@95 can be negative. Comparisons use the same minimum recall target, not exactly matched achieved recall.

Review	Method	Target positives	Screened n	Screening fraction	WSS@95	Achieved recall
Bos_2018	Jev	10	4878	1.0000	-0.0500	1.0000
Chou_2003	Jev	15	70	0.0367	0.9133	1.0000
Donners_2021	Jev	15	76	0.2946	0.6554	1.0000
Meijboom_2021	Jev	36	882	1.0000	-0.0500	1.0000
Menon_2022	Jev	71	445	0.4564	0.4936	0.9595
Nelson_2002	Jev	76	226	0.6175	0.3325	0.9500
Oud_2018	Jev	19	89	0.0935	0.8565	0.9500
Wolters_2018	Jev	19	4278	1.0000	-0.0500	1.0000
van_der_Valk_2021	Jev	85	725	1.0000	-0.0500	1.0000
van_der_Waal_2022	Jev	32	189	0.0959	0.8541	0.9697
Bos_2018	BM25	10	1903	0.3901	0.5599	1.0000
Chou_2003	BM25	15	862	0.4520	0.4980	1.0000
Donners_2021	BM25	15	249	0.9651	-0.0151	1.0000
Meijboom_2021	BM25	36	546	0.6190	0.3310	0.9730
Menon_2022	BM25	71	910	0.9333	0.0167	0.9595
Nelson_2002	BM25	76	312	0.8525	0.0975	0.9500
Oud_2018	BM25	19	389	0.4086	0.5414	0.9500
Wolters_2018	BM25	19	3555	0.8310	0.1190	1.0000
van_der_Valk_2021	BM25	85	696	0.9600	-0.0100	0.9551
van_der_Waal_2022	BM25	32	1891	0.9599	-0.0099	0.9697
Bos_2018	TF-IDF	10	698	0.1431	0.8069	1.0000
Chou_2003	TF-IDF	15	1222	0.6408	0.3092	1.0000
Donners_2021	TF-IDF	15	219	0.8488	0.1012	1.0000
Meijboom_2021	TF-IDF	36	573	0.6497	0.3003	0.9730

Review	Method	Target positives	Screened n	Screening fraction	WSS@95	Achieved recall
Menon_2022	TF-IDF	71	678	0.6954	0.2546	0.9595
Nelson_2002	TF-IDF	76	345	0.9426	0.0074	0.9500
Oud_2018	TF-IDF	19	538	0.5651	0.3849	0.9500
Wolters_2018	TF-IDF	19	1354	0.3165	0.6335	1.0000
van_der_Valk_2021	TF-IDF	85	683	0.9421	0.0079	0.9551
van_der_Waal_2022	TF-IDF	32	1593	0.8086	0.1414	0.9697
Bos_2018	MiniLM	10	1504	0.3083	0.6417	1.0000
Chou_2003	MiniLM	15	894	0.4688	0.4812	1.0000
Donners_2021	MiniLM	15	158	0.6124	0.3376	1.0000
Meijboom_2021	MiniLM	36	427	0.4841	0.4659	0.9730
Menon_2022	MiniLM	71	611	0.6267	0.3233	0.9595
Nelson_2002	MiniLM	76	260	0.7104	0.2396	0.9500
Oud_2018	MiniLM	19	440	0.4622	0.4878	0.9500
Wolters_2018	MiniLM	19	829	0.1938	0.7562	1.0000
van_der_Valk_2021	MiniLM	85	396	0.5462	0.4038	0.9551
van_der_Waal_2022	MiniLM	32	855	0.4340	0.5160	0.9697
Bos_2018	LR	10	3665	0.7513	0.1987	1.0000
Chou_2003	LR	15	1524	0.7992	0.1508	1.0000
Donners_2021	LR	15	217	0.8411	0.1089	1.0000
Meijboom_2021	LR	36	450	0.5102	0.4398	0.9730
Menon_2022	LR	71	764	0.7836	0.1664	0.9595
Nelson_2002	LR	76	338	0.9235	0.0265	0.9500
Oud_2018	LR	19	517	0.5431	0.4069	0.9500
Wolters_2018	LR	19	2877	0.6725	0.2775	1.0000
van_der_Valk_2021	LR	85	600	0.8276	0.1224	0.9551
van_der_Waal_2022	LR	32	1805	0.9162	0.0338	0.9697

Boundary ties and random-order expectations

Table S3b Boundary ties and expected screening

Source: screening_per_review.csv. Expected values average a uniformly random order within the boundary tie and may be non-integer. They do not imply discrimination within ties. Best/worst tie positions are retained in the full CSV.

Review	Method	Boundary score	Tied n	Tied inclusions	Expected screened n	Expected fraction	Expected WSS@95
Bos_2018	Jev	0.0000	4641	1	2558.0000	0.5244	0.4256
Chou_2003	Jev	0.0500	7	2	68.3333	0.0358	0.9142
Donners_2021	Jev	0.0200	7	1	73.0000	0.2829	0.6671
Meijboom_2021	Jev	0.0000	734	6	673.0000	0.7630	0.1870
Menon_2022	Jev	0.0100	137	2	400.0000	0.4103	0.5397
Nelson_2002	Jev	0.0600	7	1	223.0000	0.6093	0.3407
Oud_2018	Jev	0.0800	2	1	88.5000	0.0930	0.8570
Wolters_2018	Jev	0.0000	4150	2	2895.3333	0.6768	0.2732
van_der_Valk_2021	Jev	0.0000	569	30	634.0645	0.8746	0.0754
van_der_Waal_2022	Jev	0.0100	79	3	170.0000	0.0863	0.8637
Bos_2018	BM25	252.7825	1	1	1903.0000	0.3901	0.5599
Chou_2003	BM25	119.8505	1	1	862.0000	0.4520	0.4980
Donners_2021	BM25	15.9870	1	1	249.0000	0.9651	-0.0151
Meijboom_2021	BM25	99.0109	1	1	546.0000	0.6190	0.3310
Menon_2022	BM25	133.3977	1	1	910.0000	0.9333	0.0167
Nelson_2002	BM25	50.0885	1	1	312.0000	0.8525	0.0975
Oud_2018	BM25	95.9855	1	1	389.0000	0.4086	0.5414
Wolters_2018	BM25	127.8891	1	1	3555.0000	0.8310	0.1190
van_der_Valk_2021	BM25	38.8815	1	1	696.0000	0.9600	-0.0100
van_der_Waal_2022	BM25	101.0959	1	1	1891.0000	0.9599	-0.0099
Bos_2018	TF-IDF	0.0300	1	1	698.0000	0.1431	0.8069
Chou_2003	TF-IDF	0.0068	1	1	1222.0000	0.6408	0.3092
Donners_2021	TF-IDF	0.0025	1	1	219.0000	0.8488	0.1012
Meijboom_2021	TF-IDF	0.0068	1	1	573.0000	0.6497	0.3003

Review	Method	Boundary score	Tied n	Tied inclusions	Expected screened n	Expected fraction	Expected WSS@95
Menon_2022	TF-IDF	0.0066	1	1	678.0000	0.6954	0.2546
Nelson_2002	TF-IDF	0.0024	1	1	345.0000	0.9426	0.0074
Oud_2018	TF-IDF	0.0119	1	1	538.0000	0.5651	0.3849
Wolters_2018	TF-IDF	0.0210	1	1	1354.0000	0.3165	0.6335
van_der_Valk_2021	TF-IDF	0.0018	1	1	683.0000	0.9421	0.0079
van_der_Waal_2022	TF-IDF	0.0071	1	1	1593.0000	0.8086	0.1414
Bos_2018	MiniLM	0.5879	1	1	1504.0000	0.3083	0.6417
Chou_2003	MiniLM	0.4748	1	1	894.0000	0.4688	0.4812
Donners_2021	MiniLM	0.3383	1	1	158.0000	0.6124	0.3376
Meijboom_2021	MiniLM	0.4443	1	1	427.0000	0.4841	0.4659
Menon_2022	MiniLM	0.1992	1	1	611.0000	0.6267	0.3233
Nelson_2002	MiniLM	0.4290	1	1	260.0000	0.7104	0.2396
Oud_2018	MiniLM	0.5965	1	1	440.0000	0.4622	0.4878
Wolters_2018	MiniLM	0.4860	1	1	829.0000	0.1938	0.7562
van_der_Valk_2021	MiniLM	0.3557	1	1	396.0000	0.5462	0.4038
van_der_Waal_2022	MiniLM	0.4702	1	1	855.0000	0.4340	0.5160
Bos_2018	LR	0.1332	1	1	3665.0000	0.7513	0.1987
Chou_2003	LR	0.2314	1	1	1524.0000	0.7992	0.1508
Donners_2021	LR	0.6250	1	1	217.0000	0.8411	0.1089
Meijboom_2021	LR	0.5844	1	1	450.0000	0.5102	0.4398
Menon_2022	LR	0.1108	1	1	764.0000	0.7836	0.1664
Nelson_2002	LR	0.1910	1	1	338.0000	0.9235	0.0265
Oud_2018	LR	0.4683	1	1	517.0000	0.5431	0.4069
Wolters_2018	LR	0.1220	1	1	2877.0000	0.6725	0.2775
van_der_Valk_2021	LR	0.2344	1	1	600.0000	0.8276	0.1224
van_der_Waal_2022	LR	0.1320	1	1	1805.0000	0.9162	0.0338

Score standardization and review-level uncertainty

Table S4a Common within-review percentile scale

Source: all_models_rank_pct_metrics.csv. All five methods are converted to within-review average-rank percentiles before pooling.

Method	AUROC	AP
Jev	0.8561	0.2943
BM25	0.6657	0.0406
TF-IDF	0.6882	0.0508
MiniLM	0.8002	0.0829
LR	0.7011	0.0467

Table S4b Review-resampled macro discrimination differences

Source: review_resampling_macro_differences.csv. Ten review units; 10,000 resamples. Macro outcomes weight reviews equally. Better/worse counts do not constitute independent hypothesis tests.

Comparison	Metric	Difference	95% lower	95% upper	Reviews Jev better	Reviews Jev worse
Jev - BM25	AUROC	0.1874	0.1259	0.2517	10	0
Jev - TF-IDF	AUROC	0.1555	0.0925	0.2169	9	1
Jev - MiniLM	AUROC	0.0717	0.0293	0.1129	8	2
Jev - LR	AUROC	0.1688	0.1207	0.2137	10	0
Jev - BM25	AUPRC_AP	0.4167	0.3366	0.5027	10	0
Jev - TF-IDF	AUPRC_AP	0.3979	0.3262	0.4811	10	0
Jev - MiniLM	AUPRC_AP	0.3296	0.2487	0.4176	10	0
Jev - LR	AUPRC_AP	0.4053	0.3257	0.4911	10	0

Data-quality sensitivity analyses

Table S5 Discrimination in the archived sensitivity subsets

Source: data_quality_sensitivity.csv. Frozen predictions and original percentiles were retained. Duplicate adjustment keeps the first record within a duplicate group and removes all records in conflicting-label groups. The strict subset combines deduplication, complete text and absence of abstract-index gaps. Cross-review-overlap removal evaluates retained predictions and does not retrain LR.

Subset	Method	Records	Inclusions	AUROC	AP
primary	Jev	17191	392	0.9220	0.5012
primary	BM25	17191	392	0.6657	0.0406
primary	TF-IDF	17191	392	0.6882	0.0508
primary	MiniLM	17191	392	0.8002	0.0829
primary	LR	17191	392	0.7700	0.0848
duplicate_adjusted	Jev	17134	390	0.9222	0.5084
duplicate_adjusted	BM25	17134	390	0.6650	0.0405
duplicate_adjusted	TF-IDF	17134	390	0.6884	0.0512
duplicate_adjusted	MiniLM	17134	390	0.8007	0.0833
duplicate_adjusted	LR	17134	390	0.7698	0.0853
complete_title_abstract	Jev	16512	389	0.9217	0.5041
complete_title_abstract	BM25	16512	389	0.6575	0.0407
complete_title_abstract	TF-IDF	16512	389	0.6813	0.0518
complete_title_abstract	MiniLM	16512	389	0.7969	0.0844
complete_title_abstract	LR	16512	389	0.7664	0.0854
no_abstract_index_gap	Jev	16685	368	0.9250	0.5048
no_abstract_index_gap	BM25	16685	368	0.6719	0.0399
no_abstract_index_gap	TF-IDF	16685	368	0.6921	0.0501
no_abstract_index_gap	MiniLM	16685	368	0.8005	0.0811
no_abstract_index_gap	LR	16685	368	0.7783	0.0863
strict_clean	Jev	15964	363	0.9248	0.5150
strict_clean	BM25	15964	363	0.6636	0.0401
strict_clean	TF-IDF	15964	363	0.6854	0.0514

Subset	Method	Records	Inclusions	AUROC	AP
strict_clean	MiniLM	15964	363	0.7978	0.0829
strict_clean	LR	15964	363	0.7747	0.0875
no_cross_review_overlap	Jev	16829	387	0.9211	0.4999
no_cross_review_overlap	BM25	16829	387	0.6648	0.0406
no_cross_review_overlap	TF-IDF	16829	387	0.6856	0.0497
no_cross_review_overlap	MiniLM	16829	387	0.8012	0.0837
no_cross_review_overlap	LR	16829	387	0.7716	0.0856

Workload uncertainty across reviews

Table S6 Review-resampled WSS@95 differences

Source: screening_review_resampling.csv. Differences are Jev minus comparator, on the fraction scale. Positive values favor Jev. Resampling uses ten review units and 10,000 replicates; wide intervals reflect limited review-level information.

Tie policy	Aggregation	Comparison	Difference	95% lower	95% upper
complete_tie	record_weighted	Jev - BM25	-0.0317	-0.3537	0.4802
complete_tie	macro	Jev - BM25	0.1777	-0.1021	0.4453
complete_tie	record_weighted	Jev - TF-IDF	-0.2301	-0.6054	0.4735
complete_tie	macro	Jev - TF-IDF	0.0958	-0.2435	0.4058
complete_tie	record_weighted	Jev - MiniLM	-0.3190	-0.6091	0.2491
complete_tie	macro	Jev - MiniLM	-0.0748	-0.3652	0.2034
complete_tie	record_weighted	Jev - LR	0.0523	-0.2111	0.5377
complete_tie	macro	Jev - LR	0.1974	-0.0816	0.4694
random_tie_expectation	record_weighted	Jev - BM25	0.2053	0.0012	0.5242
random_tie_expectation	macro	Jev - BM25	0.3015	0.1122	0.4964
random_tie_expectation	record_weighted	Jev - TF-IDF	0.0070	-0.2496	0.5032
random_tie_expectation	macro	Jev - TF-IDF	0.2196	-0.0190	0.4450
random_tie_expectation	record_weighted	Jev - MiniLM	-0.0820	-0.2952	0.2772
random_tie_expectation	macro	Jev - MiniLM	0.0491	-0.1553	0.2434
random_tie_expectation	record_weighted	Jev - LR	0.2893	0.0908	0.5744
random_tie_expectation	macro	Jev - LR	0.3212	0.1152	0.5221

Probability calibration

Table S7 Calibration bins

Source: jev_calibration_bins.csv. Ten frozen equal-width bins; the last includes probability 1. Bin labels and numerical bounds are retained exactly from the analysis. No bin confidence intervals were calculated.

Bin	Lower bound	Upper bound	Records	Mean predicted probability	Observed fraction
0	0.0000	0.1000	16401	0.0019	0.0071
1	0.1000	0.2000	192	0.1357	0.1458
2	0.2000	0.3000	80	0.2391	0.1625
3	0.3000	0.4000	67	0.3476	0.1940
4	0.4000	0.5000	58	0.4467	0.2241
5	0.5000	0.6000	67	0.5425	0.3582
6	0.6000	0.7000	55	0.6491	0.2727
7	0.7000	0.8000	60	0.7527	0.4167
8	0.8000	0.9000	73	0.8463	0.6301
9	0.9000	1.0000	138	0.9522	0.7101

Total records: 17,191; final inclusions: 392. Brier score: 0.016802; Brier skill score: 0.245963; ECE: 0.012755. Exact-zero records: 15,128, including 44 inclusions. Predictions at least 0.90: n=138; mean predicted probability 0.952174; observed fraction 0.710145. Aggregate calibration error does not establish safe exclusion.

Native binary decisions by review

Table S8 Native binary decision performance

Source: per_review_metrics.csv. Jev decisions are used as returned; LR uses a 0.5 threshold. These are different operating points. TP/FN/TN/FP refer to final review inclusion, not independently labeled initial screening. Native metrics are not defined for the unthresholded retrieval baselines.

Review	Method	TP	FN	TN	FP	Sensitivity	Specificity	Precision	F1	MCC
Bos_2018	Jev	8	2	4856	12	0.8000	0.9975	0.4000	0.5333	0.5645
Chou_2003	Jev	12	3	1885	7	0.8000	0.9963	0.6316	0.7059	0.7083
Donners_2021	Jev	10	5	223	20	0.6667	0.9177	0.3333	0.4444	0.4266
Meijboom_2021	Jev	10	27	836	9	0.2703	0.9893	0.5263	0.3571	0.3585
Menon_2022	Jev	48	26	882	19	0.6486	0.9789	0.7164	0.6809	0.6570
Nelson_2002	Jev	59	21	229	57	0.7375	0.8007	0.5086	0.6020	0.4780
Oud_2018	Jev	13	7	902	30	0.6500	0.9678	0.3023	0.4127	0.4266
Wolters_2018	Jev	4	15	4257	2	0.2105	0.9995	0.6667	0.3200	0.3732
van_der_Valk_2021	Jev	35	54	617	19	0.3933	0.9701	0.6481	0.4895	0.4542
van_der_Waal_2022	Jev	9	24	1928	9	0.2727	0.9954	0.5000	0.3529	0.3616
Bos_2018	LR	3	7	4570	298	0.3000	0.9388	0.0100	0.0193	0.0449
Chou_2003	LR	14	1	1023	869	0.9333	0.5407	0.0159	0.0312	0.0840
Donners_2021	LR	15	0	27	216	1.0000	0.1111	0.0649	0.1220	0.0849
Meijboom_2021	LR	36	1	375	470	0.9730	0.4438	0.0711	0.1326	0.1689
Menon_2022	LR	4	70	864	37	0.0541	0.9589	0.0976	0.0696	0.0171
Nelson_2002	LR	61	19	95	191	0.7625	0.3322	0.2421	0.3675	0.0845
Oud_2018	LR	18	2	470	462	0.9000	0.5043	0.0375	0.0720	0.1160
Wolters_2018	LR	1	18	4160	99	0.0526	0.9768	0.0100	0.0168	0.0129
van_der_Valk_2021	LR	73	16	346	290	0.8202	0.5440	0.2011	0.3230	0.2391
van_der_Waal_2022	LR	25	8	1387	550	0.7576	0.7161	0.0435	0.0822	0.1337

Aggregated workload and integrity

Table S9 Aggregated workload

Source: screening_summary.csv. Record-weighted values aggregate screened counts over 17,191 review-record pairs. Macro values weight the ten reviews equally.

Method	Screened n	Screening fraction	WSS@95	Macro WSS@95	Random expected WSS@95	Macro random expected WSS@95	Achieved recall
BM25	11313	0.6581	0.2919	0.2128	0.2919	0.2128	0.9643
Jev	11858	0.6898	0.2602	0.3905	0.4972	0.5144	0.9770
LR	12757	0.7421	0.2079	0.1932	0.2079	0.1932	0.9643
MiniLM	6374	0.3708	0.5792	0.4653	0.5792	0.4653	0.9643
TF-IDF	7903	0.4597	0.4903	0.2947	0.4903	0.2947	0.9643

Table S10 Frozen source-file hash verification

Source: sha256_manifest.csv. Six source files had historical reference hashes; three had no independent historical reference. Current hashes are provenance records, not evidence against training contamination. Expected hashes and the full audit are in Online Resource 2.

File	Status	Current SHA256
zero_cost_baseline_protocol.json	MATCH	6e7281063ab948bf69c4154f908341c385462191046ba1a18ab271655cae539a
zero_cost_baseline_summary.txt	MATCH	39164efb19bceca194f475bbbed7ceaff2e12627985afee12a096756566035913
zero_cost_baseline_results.csv	MATCH	c663d5d1b3a2f4d56068e1dd68806ee22ebaf6e606fb287ebe1b892036c9e046
data_quality_report.csv	NO_PRIOR_HASH_AVAILABLE	c4be096545aef4a3005c9cf91f28061142e055ca8e79d2af888217168afda55f
review_context.csv	NO_PRIOR_HASH_AVAILABLE	e62f09e702a954791499e079d7f2164ee65463875d3fa59a28d22b3e7eda2484
jev_full_run_summary.txt	MATCH	778daa5ed6df7ea04f77df836ef394be89a10bc30089b11c683bb853315a463c
jev_full_protocol.json	MATCH	4565c8f92835e15285fdc87ee7132a956012cbfb5b5aece3ea32edceba5b909d
jev_full_results.csv	MATCH	b940a9a8fa9704bf1ea57ac80f1eb02d67d2d9f1c6cdbfc199a650162912ed68
jev_benchmark_master_usable.csv	NO_PRIOR_HASH_AVAILABLE	9a0d330e46ccb6ed71f5932544703495f89f5354ad489bf6ee6f9755f239358

The archived audit contains 103 checks; 103 are marked passed. Cross-review overlap consists of 362 review-record rows with 181 repeated record identifiers and five positive rows. The full identifier list is supplied for audit, not as proof that other DOI/title overlaps are absent.

Reading the numerical companion files

Versioned public frozen-record data and runnable statistical-analysis/figure code are available at <https://github.com/cx295410-dot/jev-biomedical-evidence-screening> (v1.0.0). This separate repository provides the record-level scores and quality indicators needed to recompute the results, without redistributing bibliographic text. The original API caller and baseline feature-generation programs are not included.

Online Resource 2 retains every original CSV field and numeric string, with five article-metadata columns added to each row. No values were recalculated. File-level SHA256 digests for the original and distributed files are listed in the companion inventory. The archive contains aggregate analysis outputs and overlap identifiers, not a complete record-level prediction dataset or the full inference/training pipeline.

Counts ending in `_n` are record or inclusion counts unless labeled `bootstrap_n`. Metrics and proportions use a 0-1 scale, not percent; signed differences and WSS may be negative. `ci_low` and `ci_high` are interval bounds; macro means equal review weights. Empty cells and NaN mean unavailable/inapplicable. Native binary columns are intentionally empty for retrieval methods. `score_scale` distinguishes the mixed primary pooled representation from the common percentile representation.

The companion field dictionary preserves original field names and explains key metrics and conventions. This supplement does not add inferential claims, invent missing uncertainty estimates or imply that a retrospective stopping position can be used prospectively.